

TÀI LIỆU THIẾT KẾ LỚP (CLASS DESIGN)

Hệ thống Quản lý Bãi giữ xe Thông minh

1. USER (Abstract)

«abstract» User

- **Attributes:** id: String, name: String, email: String, role: Role
- **Methods:** login(): boolean, logout(): void, getRole(): Role

Kế thừa từ User

Student: studentId: String | viewParkingHistory(): List<ParkingSession>

Lecturer: staffCode: String, department: String | viewParkingHistory(): List<ParkingSession>, getParkingBenefit(): PricingPolicy

Staff: employeeId: String, shift: String | reportIncident(desc: String): Incident, manualOverride(sessionId: String): void, confirmPayment(sessionId: String): void

Admin: configurePricing(): void, viewReports(): Report, manageUsers(): void

2. ENTITY CLASSES

1. Vehicle:

licensePlate: String, type: String
→ validate(): boolean

2. ParkingSession:

id: String, userId: String, vehicleId: String, spotId: String, cardId: String, checkInImage: String, checkInTime: DateTime, checkOutTime: DateTime, status: SessionStatus
→ startSession(): void, endSession(): void, calculateDuration(): double

3. ParkingSpot:

id: String, zoneId: String, status: SpotStatus
→ occupy(): void, release(): void

4. ParkingZone:

id: String, name: String, totalSpots: int, availableSpots: int, status: ZoneStatus
→ getAvailableSpots(): List<ParkingSpot>, updateStatus(): void

5. VisitorCard:

cardId: String, status: CardStatus
→ assignToSession(session: ParkingSession): void, reset(): void, lock(): void

6. Payment:

id: String, sessionId: String, amount: double, status: PaymentStatus, method: PaymentMethod
→ process(): boolean, validate(): boolean

7. PrepaidPackage:

id: String, name: String, durationDays: int, price: double, validFrom: DateTime, validTo: DateTime, userId: String
→ isActive(): boolean, extend(pkg: PrepaidPackage): void

8. PricingPolicy:

id: String, targetGroup: Role, pricePerHour: double, freeMinutes: int, discountPercent: double, effectiveFrom: DateTime
→ calculateFee(duration: double): double

9. Report:

id: String, type: ReportType, dateRange: DateRange, zone: String, format: String, generatedBy: String, createdAt: DateTime, content: String
→ generate(): void, export(format: String): File

10. Incident:

id: String, type: IncidentType, description: String, status: IncidentStatus, reportedBy: String, createdAt: DateTime
→ resolve(): void, notifyAdmin(): void

3. SERVICE CLASSES

- ◇ **AuthService:** loginWithSSO(token: String): User, validateUser(user: User): boolean
- ◇ **ParkingService:** createSession(cardId: String): ParkingSession, closeSession(cardId: String): void, assignSpot(): ParkingSpot, updateStatus(): void
- ◇ **PaymentService:** calculateFee(sessionId: String): double, processOnSitePayment(sessionId: String): boolean, processPrepaidPayment(sessionId: String): boolean
- ◇ **PricingService:** getCurrentPolicy(role: Role): PricingPolicy, updatePolicy(policy: PricingPolicy): void
- ◇ **ReportService:** generateReport(type: ReportType, dateRange: DateRange): Report, exportReport(id: String, format: String): File
- ◇ **IncidentService:** reportIncident(desc: String, type: IncidentType): Incident, resolveIncident(id: String): void, notifyAdmin(incident: Incident): void
- ◇ **ZoneService:** getZoneStatus(): List<ParkingZone>, updateZoneStatus(zoneId: String): void, suggestZone(role: Role): ParkingZone

4. CONTROLLER CLASSES

- **AuthController:** login(), logout()
- **ParkingController:** checkIn(), checkOut(), getZoneStatus()
- **PaymentController:** payOnSite(), payPrepaid()
- **AdminController:** configurePricing(), manageUsers(), monitorSystem()
- **ReportController:** viewReport(), exportReport()
- **IncidentController:** submitIncident(), resolveIncident()

5. EXTERNAL SYSTEM CLASSES

SSOService: authenticate(token: String): User

BKPayService: processPayment(amount: double): boolean, cancelPayment(txId: String): boolean

DataCoreService: getUser(id: String): User, syncUserList(): List<User>, checkSyncStatus(): SyncStatus

IoTGateway: status: String | sendCommand(command: String): void, receiveData(): SensorData, openBarrier(): void, getConnectionStatus(): boolean, syncOfflineData(): void

6. HARDWARE / IOT CLASSES

«abstract» Sensor

- **Attributes:** id: String, status: String | readData(): String
- **RFIDReader:** readCard(): String
- **Camera:** captureImage(): Image

Barrier

- status: String | open(): void, close(): void

7. ENUMERATIONS

Enum Name	Values
Role	STUDENT, LECTURER, STAFF, ADMIN, VISITOR
SessionStatus	ACTIVE, COMPLETED, DISPUTED
SpotStatus	AVAILABLE, OCCUPIED, RESERVED, ERROR
ZoneStatus	AVAILABLE, NEARLY_FULL, FULL
PaymentStatus	PENDING, SUCCESS, FAILED, CANCELLED
PaymentMethod	CASH, QR, PREPAID
CardStatus	AVAILABLE, IN_USE, LOCKED
IncidentType	HARDWARE, ON_SITE
IncidentStatus	PENDING, IN_PROGRESS, RESOLVED
ReportType	REVENUE, USAGE, FREQUENCY